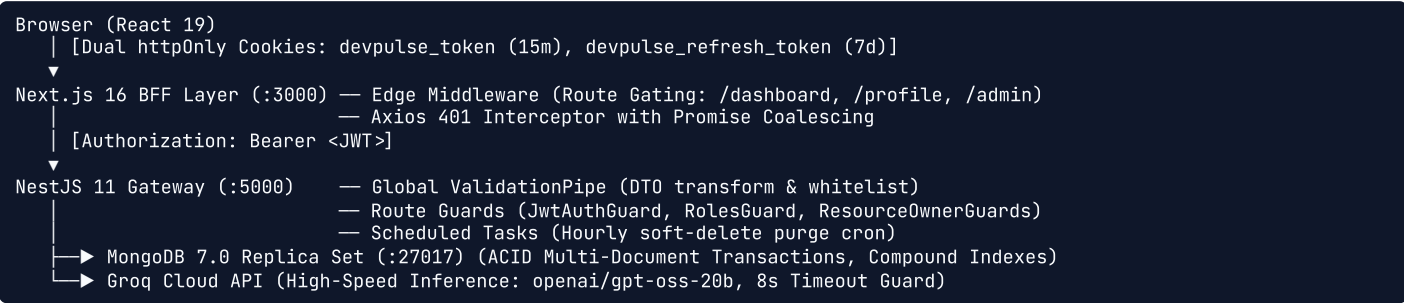


- NESTJS 11
- NEXT.JS 16 (APP ROUTER)
- MONGODB 7.0 REPLICA SET
- DOCKER COMPOSE
- GROQ AI (GPT-OSS-20B)

1. End-to-End System Topology & Request Lifecycle

DevPulse implements the **Backend-For-Frontend (BFF)** architectural pattern. The browser interacts exclusively with Next.js App Router route handlers, which act as a secure proxy forwarding authenticated requests to NestJS.



2. Authentication & Session Rotation Workflow

1. Signup & Login Flow

User submits credentials via RHF + Zod (onTouched mode). Backend verifies password via bcrypt (10 salt rounds), issues 15-minute Access Token and 7-day Refresh Token. Next.js BFF sets dual httpOnly , SameSite=Lax , Secure cookies.

2. SHA-256 Refresh Token Rotation

Raw refresh tokens are never stored in MongoDB. The database stores only deterministic SHA-256 digest hashes (User.refreshTokenHash). During token rotation, MongoDB performs atomic compare-and-swap rotation, immediately invalidating old tokens.

3. Core Community Workflows

Post Authoring & Feed Ranking Algorithm

GET /posts?sort=top|latest|most-discussed

Posts support full Markdown with syntax-highlighted code blocks. Feeds are sorted deterministically:

rankScore = (likes - dislikes) + (commentCount * 2) [Secondary tie-breaker: createdAt DESC, _id DESC]

TanStack Query isolates feed caches by sort key (['posts', 'feed', sort]), enabling instantaneous zero-flicker tab switching.

Threaded & Flattened Comments with @Mentions

POST /posts/:id/comments/:cId/replies

Constrains nesting to depth = 1 to prevent mobile layout squishing. When replying to an existing reply, the system captures mentionedUserId and renders an @DeveLoper badge at the same indentation level. Deleting root comments atomically cascades to all replies inside a MongoDB transaction.

4. Architectural Decisions: Backend & Infrastructure

1. MongoDB ACID Transactions for Reactions

Context: Concurrency race conditions on rapid like/dislike clicks caused post counter drift.

Decision: Wrap toggle operations in MongoDB replica-set transactions (`ClientSession`) with an automated retry loop.

✓ **Advantage:** 100% strict consistency between counter pills and reaction records.

✗ **Trade-off:** Mandates a MongoDB Replica Set (`rs0`) in Docker/local environments.

2. Lean Author Projection vs. Document Bloat

Context: Populating complete User documents bloated feed payloads with sensitive hashes and heavy subdocuments.

Decision: Enforced strict projection: `.populate({ path: 'authorId', select: 'name headline avatarUrl' })` .

✓ **Advantage:** Slashes payload size by >70%; zero risk of leaking security hashes.

✗ **Trade-off:** Extended profile views require a secondary call to `/users/:id` .

3. Depth-1 Flattened Reply Architecture

Context: Recursive comment trees caused mobile UI horizontal squishing and recursive database aggregation queries.

Decision: Limit nesting depth to 1 and store `mentionedUserId` for conversational tagging.

✓ **Advantage:** Responsive ergonomics on narrow mobile screens (≤375px); simple queries.

✗ **Trade-off:** Users cannot branch deeply nested multi-level conversation trees.

4. Groq LLM Inference with Boundary Guards

Context: Summarizing long posts with AI can cause latency spikes, prompt bloat, and API failures.

Decision: 12k character body truncation, 8-second timeout guard, and fallback to `MockSummarizerProvider` .

✓ **Advantage:** Sub-1.5s inference time; offline/CI environments never break.

✗ **Trade-off:** Extremely long articles (>12,000 chars) are summarized based on initial text.

5. Architectural Decisions: Frontend & Client State

5. Shared-Promise 401 Interceptor Coalescing

Context: Simultaneous queries failing with 401 triggered parallel refresh calls, invalidating rotated tokens.

Decision: Module-scoped singleton `refreshPromise` coalesces all concurrent 401s into one request.

✓ **Advantage:** Zero token collisions; seamless background replay without user interruption.

✗ **Trade-off:** Requires careful memory management to prevent hanging promises on dropouts.

6. Sort-Aware Query-Key Cache Partitioning

Context: Toggling between feed tabs (Top, Latest, Discussed) previously triggered refetch flickers.

Decision: Partition query keys: `['posts', 'feed', sort, { limit }]` synced with URL `?sort=` .

✓ **Advantage:** Instantaneous tab switching directly from TanStack Query memory.

✗ **Trade-off:** Retains up to three paginated lists concurrently in client memory.

7. Optimistic Reaction Mutations with Rollback

Context: Like/dislike actions need instant tactile response without waiting for network round-trips.

Decision: Use TanStack Query `onMutate` snapshotting with atomic rollback in `onError` .

✓ **Advantage:** Instant UI feedback (<16ms) with animated count transitions.

✗ **Trade-off:** Requires manual counter and pill state rollback logic on failed requests.

8. Dual httpOnly Cookies via Next.js BFF

Context: Storing JWT in browser localStorage exposes tokens to XSS exfiltration.

Decision: Route all auth through Next.js BFF handlers with `httpOnly` cookies and Edge middleware.

✓ **Advantage:** Immune to JavaScript XSS access; Edge route gating before layout rendering.

✗ **Trade-off:** Extra network hop through Next.js server proxy for client API mutations.

6. Complete 20-Day Milestone Execution Matrix

Phase	Days	Milestones & Core Capabilities	Key Engineering Deliverables	Status
Phase 1	Days 1–4	Foundations, API Contracts & Authentication	NestJS modular layout, Mongoose setup, /health probe, Swagger /docs, TransformInterceptor, HttpExceptionHandler, bcrypt, JWT access/refresh, httpOnly cookie BFF, RHF+Zod, Edge middleware.	✓ Completed
Phase 2	Days 5–8	Developer Profiles, Forms & Post Feed	Headline/bio, nested skills, work experiences, portfolio projects with date validation, useFieldArray, Post schema, soft-delete lifecycle (5-day restore), hourly cleanup cron, infinite feed.	✓ Completed
Phase 3	Days 9–12	Discussions, Concurrency & Optimistic UI	Threaded comments, depth-1 reply enforcement, @Mentions, reaction engine with MongoDB transactions, optimistic mutations, reactors list modal with tabs, hover peek popovers, clickable profiles.	✓ Completed
Phase 4	Days 13–16	Discovery, Ranking Engines & AI Summarizer	Multi-criteria feed sorting (top, latest, most-discussed), deterministic rank score pipeline, compound indexes, debounced full-text search, Groq Cloud LLM (openai/gpt-oss-20b), 8s timeout guard, mock fallback.	✓ Completed
Phase 5	Days 17–20	Testing, Hardening, DevOps & Final Demo	274-test full-stack test matrix (212 backend, 62 frontend), SHA-256 token rotation, Throttler rate limits, multi-stage Dockerfiles, Docker Compose replica set rs0, Day 20 smoke tests, complete docs.	✓ Completed

7. Custom Features Beyond Original Plan (Explicit Highlights)

1. Clickable Commenter Profiles

/developers/[id]

Commenter avatars & names link directly to public developer portfolios, backed by lean projection queries that never leak credentials.

2. Flattened Same-Depth Replies

depth = 1

Replies render at level 1 with structured @Mention tags referencing mentionedUserId , preventing mobile UI compression.

3. Reactors Modal & Hover Peek

GET /reactions

300ms hover popover avatar stack plus paginated modal dialog filterable by All, Likes (👍), and Dislikes (👎).

4. Groq Cloud AI Summarizer

gpt-oss-20b

On-demand bullet summaries & extracted skill tags with 12k char truncation, 8s timeout guard, and mock fallback.

8. Pre-Demo Rehearsal & Verification Checklist

Environment & Primary Flows:

- [x] Docker stack up: docker compose up --build -d (healthy)
- [x] App live on :3000 , Swagger live on :5000/docs
- [x] Register & Login with dual httpOnly cookies
- [x] Toggle Top / Latest / Discussed feeds (instant switch)
- [x] Search with debounced text relevance ranking
- [x] Create post with Markdown code snippets

Custom Features & Resilience Demo:

- [x] Click commenter avatar → navigates to profile
- [x] Post flattened reply with automatic @Mention
- [x] Toggle like & inspect Reactors Modal tabs
- [x] Click "Summarize Post" (Groq AI bullets & tags)
- [x] Admin access control: Non-admin gets 403 Forbidden
- [x] **Failure Showcase:** Delete devpulse_token → silent 401 refresh